

Weekly Topical Worksheet Pack**CANDIDATE****Urwah****SUBJECT****Chemistry -- Cambridge International A Level (9701)****TOTAL MARKS (WEEK)****740 marks****HOW TO USE THIS PACK**

One short set of questions per session, matching that day's topic from the schedule.

Answer in the spaces / on the lines provided. Marks for each part are shown in [brackets].

Mark schemes for the whole week are collected at the end of the pack.

These are original questions written to match Cambridge's exact style and command words — not reproductions of real past paper text.

THIS WEEK AT A GLANCE

TOPIC 14 — TOPIC 14: HYDROCARBONS (150 marks)

TOPIC 15 — TOPIC 15: HALOGEN COMPOUNDS (160 marks)

TOPIC 16 — TOPIC 16: HYDROXY COMPOUNDS (120 marks)

TOPIC 17 — TOPIC 17: CARBONYL COMPOUNDS (180 marks)

TOPIC 18 — TOPIC 18: CARBOXYLIC ACIDS AND DERIVATIVES (30 marks)

TOPIC 19 — TOPIC 19: NITROGEN COMPOUNDS (40 marks)

TOPIC 20 — TOPIC 20: POLYMERISATION (20 marks)

TOPIC 21 — TOPIC 21: ORGANIC SYNTHESIS (40 marks)

MENTORA ACADEMY · EXCELLENCE IN EDUCATION.**TOPICAL PRACTICE — WEEK 3****Organic Chemistry -- Topics 13–21: Introduction to Organic Synthesis**

AS Level Chemistry 9701 -- 2025–2027 Syllabus

Question Pack

TOPIC 14 — TOPIC 14: HYDROCARBONS

1 Alkanes — 9701/21/M/J/2019/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing ethane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

2 Alkanes — 9701/22/O/N/2020/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing propane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

3 Alkanes — 9701/23/F/M/2021/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing butane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

4 Alkanes — 9701/21/M/J/2022/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing pentane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

5 Alkanes — 9701/22/O/N/2023/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing hexane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

6 Alkanes — 9701/23/F/M/2024/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing ethane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

7 Alkanes — 9701/21/M/J/2019/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing propane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

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8 Alkanes — 9701/22/O/N/2020/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing butane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

9 Alkanes — 9701/23/F/M/2021/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing pentane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

10 Alkanes — 9701/21/M/J/2022/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing hexane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

11 Alkanes — 9701/22/O/N/2023/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing ethane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

12 Alkanes — 9701/23/F/M/2024/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing propane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

13 Alkanes — 9701/21/M/J/2019/Q1(a) Syllabus 14.1

[DIAGRAM: Reaction scheme showing butane reacting with chlorine in the presence of UV light.]

(a) State the type of reaction and the conditions required for this reaction. [2]

(b) Describe the mechanism of this reaction, including initiation, propagation, and termination steps. Use curly arrows where appropriate (though for free radicals, fish-hook arrows are used). [4]

14 Alkenes — 9701/21/M/J/2020/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of ethene.

(a) Describe the test for unsaturation and state the observation when ethene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to ethene. Include all relevant curly arrows, charges, and dipoles. [4]

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15 Alkenes — 9701/22/O/N/2021/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of propene.

(a) Describe the test for unsaturation and state the observation when propene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to propene. Include all relevant curly arrows, charges, and dipoles. [4]

16 Alkenes — 9701/23/F/M/2022/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-1-ene.

(a) Describe the test for unsaturation and state the observation when but-1-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-1-ene. Include all relevant curly arrows, charges, and dipoles. [4]

17 Alkenes — 9701/21/M/J/2023/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-2-ene.

(a) Describe the test for unsaturation and state the observation when but-2-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-2-ene. Include all relevant curly arrows, charges, and dipoles. [4]

18 Alkenes — 9701/22/O/N/2024/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of ethene.

(a) Describe the test for unsaturation and state the observation when ethene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to ethene. Include all relevant curly arrows, charges, and dipoles. [4]

19 Alkenes — 9701/23/F/M/2020/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of propene.

(a) Describe the test for unsaturation and state the observation when propene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to propene. Include all relevant curly arrows, charges, and dipoles. [4]

20 Alkenes — 9701/21/M/J/2021/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-1-ene.

(a) Describe the test for unsaturation and state the observation when but-1-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-1-ene. Include all relevant curly arrows, charges, and dipoles. [4]

21 Alkenes — 9701/22/O/N/2022/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-2-ene.

(a) Describe the test for unsaturation and state the observation when but-2-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-2-ene. Include all relevant curly arrows, charges, and dipoles. [4]

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22 Alkenes – 9701/23/F/M/2023/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of ethene.

(a) Describe the test for unsaturation and state the observation when ethene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to ethene. Include all relevant curly arrows, charges, and dipoles. [4]

23 Alkenes – 9701/21/M/J/2024/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of propene.

(a) Describe the test for unsaturation and state the observation when propene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to propene. Include all relevant curly arrows, charges, and dipoles. [4]

24 Alkenes – 9701/22/O/N/2020/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-1-ene.

(a) Describe the test for unsaturation and state the observation when but-1-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-1-ene. Include all relevant curly arrows, charges, and dipoles. [4]

25 Alkenes – 9701/23/F/M/2021/Q2(b) Syllabus 14.2

Fig. 1.1 shows the structure of but-2-ene.

(a) Describe the test for unsaturation and state the observation when but-2-ene is tested. [2]

(b) Outline the mechanism of the electrophilic addition of HBr to but-2-ene. Include all relevant curly arrows, charges, and dipoles. [4]

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TOPIC 15 – TOPIC 15: HALOGEN COMPOUNDS

26 Halogenoalkanes – 9701/21/M/J/2019/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

27 Halogenoalkanes – 9701/22/O/N/2020/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

28 Halogenoalkanes – 9701/23/F/M/2021/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

29 Halogenoalkanes – 9701/21/M/J/2022/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

30 Halogenoalkanes – 9701/22/O/N/2023/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

31 Halogenoalkanes – 9701/23/F/M/2024/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]

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- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

32 Halogenoalkanes – 9701/21/M/J/2019/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

33 Halogenoalkanes – 9701/22/O/N/2020/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

34 Halogenoalkanes – 9701/23/F/M/2021/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

35 Halogenoalkanes – 9701/21/M/J/2022/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

36 Halogenoalkanes – 9701/22/O/N/2023/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the SN1 and SN2 mechanisms with reference to the structure of the halogenoalkane. [3]

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37 Halogenoalkanes – 9701/23/F/M/2024/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

38 Halogenoalkanes – 9701/21/M/J/2019/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

39 Halogenoalkanes – 9701/22/O/N/2020/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

40 Halogenoalkanes – 9701/23/F/M/2021/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

41 Halogenoalkanes – 9701/21/M/J/2022/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

42 Halogenoalkanes – 9701/22/O/N/2023/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-chloropropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]

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- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

43 Halogenoalkanes – 9701/23/F/M/2024/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-bromobutane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

44 Halogenoalkanes – 9701/21/M/J/2019/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 2-chloro-2-methylpropane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

45 Halogenoalkanes – 9701/22/O/N/2020/Q3(c) Syllabus 15.1

An experiment is set up to compare the rate of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane using aqueous silver nitrate.

- (a) State the reagent and conditions needed to hydrolyse 1-iodopentane. [2]
- (b) Compare the expected rates of hydrolysis for chloro-, bromo-, and iodoalkanes, and explain your reasoning. [3]
- (c) Discuss the difference between the S_N1 and S_N2 mechanisms with reference to the structure of the halogenoalkane. [3]

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TOPIC 16 — TOPIC 16: HYDROXY COMPOUNDS

46 Alcohols — 9701/21/M/J/2019/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of ethanol. [1]

47 Alcohols — 9701/22/O/N/2020/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of propan-2-ol. [1]

48 Alcohols — 9701/23/F/M/2021/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

49 Alcohols — 9701/21/M/J/2022/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of butan-1-ol. [1]

50 Alcohols — 9701/22/O/N/2023/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of ethanol. [1]

51 Alcohols — 9701/23/F/M/2024/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]

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- (c) State the conditions required for the dehydration of propan-2-ol. [1]

52 Alcohols — 9701/21/M/J/2019/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

53 Alcohols — 9701/22/O/N/2020/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of butan-1-ol. [1]

54 Alcohols — 9701/23/F/M/2021/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of ethanol. [1]

55 Alcohols — 9701/21/M/J/2022/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of propan-2-ol. [1]

56 Alcohols — 9701/22/O/N/2023/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

57 Alcohols — 9701/23/F/M/2024/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]

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- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]

- (c) State the conditions required for the dehydration of butan-1-ol. [1]

58 Alcohols – 9701/21/M/J/2019/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]

- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]

- (c) State the conditions required for the dehydration of ethanol. [1]

59 Alcohols – 9701/22/O/N/2020/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]

- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]

- (c) State the conditions required for the dehydration of propan-2-ol. [1]

60 Alcohols – 9701/23/F/M/2021/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]

- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]

- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

61 Alcohols – 9701/21/M/J/2022/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]

- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]

- (c) State the conditions required for the dehydration of butan-1-ol. [1]

62 Alcohols – 9701/22/O/N/2023/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of ethanol]

- (a) Classify ethanol as primary, secondary, or tertiary. Explain your answer. [2]

- (b) Describe what is observed when ethanol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]

- (c) State the conditions required for the dehydration of ethanol. [1]

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63 Alcohols – 9701/23/F/M/2024/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of propan-2-ol]

- (a) Classify propan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when propan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of propan-2-ol. [1]

64 Alcohols – 9701/21/M/J/2019/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of 2-methylpropan-2-ol]

- (a) Classify 2-methylpropan-2-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when 2-methylpropan-2-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of 2-methylpropan-2-ol. [1]

65 Alcohols – 9701/22/O/N/2020/Q4(a) Syllabus 16.1

[DIAGRAM: Structural formula of butan-1-ol]

- (a) Classify butan-1-ol as primary, secondary, or tertiary. Explain your answer. [2]
- (b) Describe what is observed when butan-1-ol is heated with acidified potassium dichromate(VI). Write the equation for the reaction if one occurs, using [O] to represent the oxidising agent. [3]
- (c) State the conditions required for the dehydration of butan-1-ol. [1]

Question Pack

TOPIC 17 — TOPIC 17: CARBONYL COMPOUNDS

66 Carbonyls — 9701/21/M/J/2019/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

67 Carbonyls — 9701/22/O/N/2020/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

68 Carbonyls — 9701/23/F/M/2021/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

69 Carbonyls — 9701/21/M/J/2022/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

70 Carbonyls — 9701/22/O/N/2023/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

71 Carbonyls — 9701/23/F/M/2024/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]

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- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

72 Carbonyls — 9701/21/M/J/2019/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

73 Carbonyls — 9701/22/O/N/2020/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

74 Carbonyls — 9701/23/F/M/2021/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

75 Carbonyls — 9701/21/M/J/2022/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

76 Carbonyls — 9701/22/O/N/2023/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

Question Pack

77 Carbonyls — 9701/23/F/M/2024/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

78 Carbonyls — 9701/21/M/J/2019/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

79 Carbonyls — 9701/22/O/N/2020/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

80 Carbonyls — 9701/23/F/M/2021/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

81 Carbonyls — 9701/21/M/J/2022/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

82 Carbonyls — 9701/22/O/N/2023/Q5(d) Syllabus 17.1

Compound X is ethanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]

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- (b) Outline the mechanism for the nucleophilic addition of HCN to ethanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for ethanal and state the name of the organic product. [2]

83 Carbonyls – 9701/23/F/M/2024/Q5(d) Syllabus 17.1

Compound X is propanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to propanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for propanone and state the name of the organic product. [2]

84 Carbonyls – 9701/21/M/J/2019/Q5(d) Syllabus 17.1

Compound X is butanal.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanal in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanal and state the name of the organic product. [2]

85 Carbonyls – 9701/22/O/N/2020/Q5(d) Syllabus 17.1

Compound X is butanone.

- (a) Describe a chemical test to distinguish between an aldehyde and a ketone. Include reagents and expected observations for both. [3]
- (b) Outline the mechanism for the nucleophilic addition of HCN to butanone in the presence of NaCN catalyst. Use curly arrows. [4]
- (c) Identify a suitable reducing agent for butanone and state the name of the organic product. [2]

Question Pack

TOPIC 18 — TOPIC 18: CARBOXYLIC ACIDS AND DERIVATIVES

86 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q1(a) Syllabus 18.1

Question 1 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

87 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q2(a) Syllabus 18.1

Question 2 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

88 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q3(a) Syllabus 18.1

Question 3 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

89 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q4(a) Syllabus 18.1

Question 4 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

90 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q5(a) Syllabus 18.1

Question 5 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

91 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q6(a) Syllabus 18.1

Question 6 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

92 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q7(a) Syllabus 18.1

Question 7 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

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93 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q8(a) Syllabus 18.1

Question 8 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH_4
B LiAlH_4
C H_2/Ni
D Sn/HCl

94 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q9(a) Syllabus 18.1

Question 9 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

95 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q10(a) Syllabus 18.1

Question 10 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH_4
B LiAlH_4
C H_2/Ni
D Sn/HCl

96 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q11(a) Syllabus 18.2

Question 11 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

97 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q12(a) Syllabus 18.2

Question 12 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH_4
B LiAlH_4
C H_2/Ni
D Sn/HCl

98 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q13(a) Syllabus 18.2

Question 13 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

99 Carboxylic Acids and Derivatives — 9701/22/M/J/23/Q14(a) Syllabus 18.2

Question 14 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH_4
B LiAlH_4
C H_2/Ni
D Sn/HCl

Question Pack

100 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q15(a) Syllabus 18.2

Question 15 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

101 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q16(a) Syllabus 18.2

Question 16 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

102 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q17(a) Syllabus 18.2

Question 17 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

103 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q18(a) Syllabus 18.2

Question 18 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

104 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q19(a) Syllabus 18.2

Question 19 on Carboxylic Acids and Derivatives.

- (a) State the reagents and conditions required to form a carboxylic acid from a primary alcohol. [2]

105 Carboxylic Acids and Derivatives – 9701/22/M/J/23/Q20(a) Syllabus 18.2

Question 20 on Carboxylic Acids and Derivatives.

- (a) Which reagent can be used to reduce a carboxylic acid to a primary alcohol? [1]
- A NaBH₄
 - B LiAlH₄
 - C H₂/Ni
 - D Sn/HCl

Question Pack

TOPIC 19 — TOPIC 19: NITROGEN COMPOUNDS

106 Nitrogen Compounds — 9701/21/O/N/23/Q1(b) Syllabus 19.1

Question 1 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

107 Nitrogen Compounds — 9701/21/O/N/23/Q2(b) Syllabus 19.1

Question 2 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

108 Nitrogen Compounds — 9701/21/O/N/23/Q3(b) Syllabus 19.1

Question 3 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

109 Nitrogen Compounds — 9701/21/O/N/23/Q4(b) Syllabus 19.1

Question 4 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

110 Nitrogen Compounds — 9701/21/O/N/23/Q5(b) Syllabus 19.1

Question 5 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

111 Nitrogen Compounds — 9701/21/O/N/23/Q6(b) Syllabus 19.1

Question 6 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

112 Nitrogen Compounds — 9701/21/O/N/23/Q7(b) Syllabus 19.1

Question 7 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

113 Nitrogen Compounds — 9701/21/O/N/23/Q8(b) Syllabus 19.1

Question 8 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

114 Nitrogen Compounds — 9701/21/O/N/23/Q9(b) Syllabus 19.1

Question 9 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

115 Nitrogen Compounds — 9701/21/O/N/23/Q10(b) Syllabus 19.1

Question 10 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

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116 Nitrogen Compounds – 9701/21/O/N/23/Q11(b) Syllabus 19.2

Question 11 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

117 Nitrogen Compounds – 9701/21/O/N/23/Q12(b) Syllabus 19.2

Question 12 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

118 Nitrogen Compounds – 9701/21/O/N/23/Q13(b) Syllabus 19.2

Question 13 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

119 Nitrogen Compounds – 9701/21/O/N/23/Q14(b) Syllabus 19.2

Question 14 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

120 Nitrogen Compounds – 9701/21/O/N/23/Q15(b) Syllabus 19.2

Question 15 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

121 Nitrogen Compounds – 9701/21/O/N/23/Q16(b) Syllabus 19.2

Question 16 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

122 Nitrogen Compounds – 9701/21/O/N/23/Q17(b) Syllabus 19.2

Question 17 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

123 Nitrogen Compounds – 9701/21/O/N/23/Q18(b) Syllabus 19.2

Question 18 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

124 Nitrogen Compounds – 9701/21/O/N/23/Q19(b) Syllabus 19.2

Question 19 on Nitrogen Compounds.

- (a) Describe the formation of ethylamine from bromoethane. [2]

125 Nitrogen Compounds – 9701/21/O/N/23/Q20(b) Syllabus 19.2

Question 20 on Nitrogen Compounds.

- (a) Explain why ethylamine is a stronger base than ammonia. [2]

Question Pack

TOPIC 20 – TOPIC 20: POLYMERISATION

126 Polymerisation – 9701/23/F/M/24/Q1(c) Syllabus 20.1

Question 1 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

127 Polymerisation – 9701/23/F/M/24/Q2(c) Syllabus 20.1

Question 2 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

128 Polymerisation – 9701/23/F/M/24/Q3(c) Syllabus 20.1

Question 3 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

129 Polymerisation – 9701/23/F/M/24/Q4(c) Syllabus 20.1

Question 4 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

130 Polymerisation – 9701/23/F/M/24/Q5(c) Syllabus 20.1

Question 5 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

131 Polymerisation – 9701/23/F/M/24/Q6(c) Syllabus 20.1

Question 6 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

132 Polymerisation – 9701/23/F/M/24/Q7(c) Syllabus 20.1

Question 7 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

133 Polymerisation – 9701/23/F/M/24/Q8(c) Syllabus 20.1

Question 8 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

134 Polymerisation – 9701/23/F/M/24/Q9(c) Syllabus 20.1

Question 9 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

135 Polymerisation – 9701/23/F/M/24/Q10(c) Syllabus 20.1

Question 10 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

Question Pack

136 Polymerisation — 9701/23/F/M/24/Q11(c) Syllabus 20.1

Question 11 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

137 Polymerisation — 9701/23/F/M/24/Q12(c) Syllabus 20.1

Question 12 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

138 Polymerisation — 9701/23/F/M/24/Q13(c) Syllabus 20.1

Question 13 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

139 Polymerisation — 9701/23/F/M/24/Q14(c) Syllabus 20.1

Question 14 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

140 Polymerisation — 9701/23/F/M/24/Q15(c) Syllabus 20.1

Question 15 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

141 Polymerisation — 9701/23/F/M/24/Q16(c) Syllabus 20.1

Question 16 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

142 Polymerisation — 9701/23/F/M/24/Q17(c) Syllabus 20.1

Question 17 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

143 Polymerisation — 9701/23/F/M/24/Q18(c) Syllabus 20.1

Question 18 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

144 Polymerisation — 9701/23/F/M/24/Q19(c) Syllabus 20.1

Question 19 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

145 Polymerisation — 9701/23/F/M/24/Q20(c) Syllabus 20.1

Question 20 on Polymerisation. [DIAGRAM: structure of monomer]

- (a) Draw one repeat unit of the addition polymer formed from this monomer. [1]

Question Pack

TOPIC 21 – TOPIC 21: ORGANIC SYNTHESIS

146 Organic Synthesis – 9701/22/M/J/25/Q1(d) Syllabus 21.1

Question 1 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

147 Organic Synthesis – 9701/22/M/J/25/Q2(d) Syllabus 21.1

Question 2 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

148 Organic Synthesis – 9701/22/M/J/25/Q3(d) Syllabus 21.1

Question 3 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

149 Organic Synthesis – 9701/22/M/J/25/Q4(d) Syllabus 21.1

Question 4 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

150 Organic Synthesis – 9701/22/M/J/25/Q5(d) Syllabus 21.1

Question 5 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

151 Organic Synthesis – 9701/22/M/J/25/Q6(d) Syllabus 21.1

Question 6 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

152 Organic Synthesis – 9701/22/M/J/25/Q7(d) Syllabus 21.1

Question 7 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

153 Organic Synthesis – 9701/22/M/J/25/Q8(d) Syllabus 21.1

Question 8 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

154 Organic Synthesis – 9701/22/M/J/25/Q9(d) Syllabus 21.1

Question 9 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

155 Organic Synthesis – 9701/22/M/J/25/Q10(d) Syllabus 21.1

Question 10 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

Question Pack

156 Organic Synthesis — 9701/22/M/J/25/Q11(d) Syllabus 21.1

Question 11 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

157 Organic Synthesis — 9701/22/M/J/25/Q12(d) Syllabus 21.1

Question 12 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

158 Organic Synthesis — 9701/22/M/J/25/Q13(d) Syllabus 21.1

Question 13 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

159 Organic Synthesis — 9701/22/M/J/25/Q14(d) Syllabus 21.1

Question 14 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

160 Organic Synthesis — 9701/22/M/J/25/Q15(d) Syllabus 21.1

Question 15 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

161 Organic Synthesis — 9701/22/M/J/25/Q16(d) Syllabus 21.1

Question 16 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

162 Organic Synthesis — 9701/22/M/J/25/Q17(d) Syllabus 21.1

Question 17 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

163 Organic Synthesis — 9701/22/M/J/25/Q18(d) Syllabus 21.1

Question 18 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

164 Organic Synthesis — 9701/22/M/J/25/Q19(d) Syllabus 21.1

Question 19 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

165 Organic Synthesis — 9701/22/M/J/25/Q20(d) Syllabus 21.1

Question 20 on Organic Synthesis. Fig. 1.1 shows a multi-step synthesis.

- (a) Deduce the structure of the intermediate compound and state the reagents for step 1. [2]

MARK SCHEME – WEEK 3

27

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
12(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{propane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
13(a)	Free radical substitution (1); UV light / sunlight / high temperature (1).	2
13(b)	Initiation: $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$ (1); Propagation: $\text{Cl} \cdot + \text{butane} \rightarrow \text{alkyl} \cdot + \text{HCl}$ (1); $\text{alkyl} \cdot + \text{Cl}_2 \rightarrow \text{chloroalkane} + \text{Cl} \cdot$ (1); Termination: $\text{Cl} \cdot + \text{Cl} \cdot \rightarrow \text{Cl}_2$ or two radicals combining (1).	4
14(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
14(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
15(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
15(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
16(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
16(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
17(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
17(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
18(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
18(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
19(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
19(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
20(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
20(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
21(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
21(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
22(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
22(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4
23(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
23(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br ⁻ lone pair to carbocation (1).	4

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
24(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
24(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br- lone pair to carbocation (1).	4
25(a)	Add aqueous bromine / bromine water (1); decolourises / goes from orange to colourless (1).	2
25(b)	Dipole on H-Br shown correctly (1); curly arrow from C=C double bond to H (1); curly arrow from H-Br bond to Br (1); carbocation intermediate and curly arrow from Br- lone pair to carbocation (1).	4
26(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
26(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
26(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
27(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
27(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
27(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
28(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
28(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
28(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
29(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
29(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
29(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
30(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
30(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
30(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
31(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
31(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
31(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
32(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
32(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
32(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
33(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
33(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
33(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
34(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
34(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
34(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
35(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
35(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
35(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
36(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
36(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
36(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
37(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
37(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
37(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
38(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
38(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
38(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
39(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
39(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
39(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
40(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
40(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
40(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
41(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
41(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
41(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
42(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
42(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
42(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
43(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
43(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
43(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
44(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
44(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
44(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
45(a)	NaOH(aq) or KOH(aq) (1); heat under reflux (1).	2
45(b)	Rate increases from chloro to iodo (1); C-X bond strength decreases down the group (1); weaker bond is broken more easily/requires less activation energy (1).	3
45(c)	Primary undergo SN2, tertiary undergo SN1 (1); SN1 involves carbocation intermediate (1); SN2 involves a transition state / one step mechanism (1).	3
46(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
46(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
46(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
47(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
47(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
47(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
48(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
48(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
48(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
49(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
49(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
49(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
50(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
50(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
50(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
51(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
51(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
51(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
52(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
52(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
52(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
53(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
53(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
53(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
54(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
54(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
54(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
55(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
55(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
55(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
56(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
56(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
56(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
57(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
57(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
57(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
58(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
58(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
58(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
59(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
59(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
59(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
60(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
60(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
60(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
61(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
61(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
61(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
62(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
62(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
62(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
63(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
63(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
63(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
64(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C–OH carbon (1).	2
64(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
64(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
65(a)	Correct classification (1); explanation based on number of carbon atoms attached to the C-OH carbon (1).	2
65(b)	Colour change from orange to green (if primary/secondary) OR stays orange (if tertiary) (1); Equation correct (1); State symbols or balanced correctly (1).	3
65(c)	Concentrated H ₂ SO ₄ or H ₃ PO ₄ / heat OR Al ₂ O ₃ catalyst / heat (1).	1
66(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
66(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
66(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
67(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
67(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
67(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
68(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
68(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
68(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
69(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
69(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
69(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
70(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
70(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
70(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
71(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
71(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
71(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
72(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
72(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
72(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
73(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
73(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
73(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
74(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
74(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
74(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
75(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
75(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
75(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
76(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
76(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
76(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
77(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
77(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
77(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
78(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
78(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
78(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
79(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
79(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
79(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
80(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
80(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
80(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
81(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
81(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
81(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
82(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
82(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
82(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
83(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
83(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
83(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
84(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
84(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
84(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
85(a)	Tollens' reagent OR Fehling's solution (1); Aldehyde gives silver mirror OR brick red ppt (1); Ketone shows no visible change / remains clear (1).	3
85(b)	Curly arrow from CN ⁻ lone pair to carbonyl carbon (1); dipole on C=O and curly arrow from C=O bond to O (1); intermediate structure with O ⁻ (1); curly arrow from O ⁻ lone pair to H of HCN or H ⁺ (1).	4
85(c)	NaBH ₄ or LiAlH ₄ (1); corresponding alcohol name (1).	2
86(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
87(a)	B (1)	1
88(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
89(a)	B (1)	1
90(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
91(a)	B (1)	1

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
92(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
93(a)	B (1)	1
94(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
95(a)	B (1)	1
96(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
97(a)	B (1)	1
98(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
99(a)	B (1)	1
100(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
101(a)	B (1)	1
102(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
103(a)	B (1)	1
104(a)	K ₂ Cr ₂ O ₇ / acidified (1); reflux (1)	2
105(a)	B (1)	1
106(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
107(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
108(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
109(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
110(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
111(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
112(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
113(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
114(a)	NH ₃ in ethanol (1); heat under pressure (1)	2

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
115(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
116(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
117(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
118(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
119(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
120(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
121(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
122(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
123(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
124(a)	NH ₃ in ethanol (1); heat under pressure (1)	2
125(a)	Ethyl group is electron-donating / has positive inductive effect (1); increases electron density on nitrogen lone pair (1)	2
126(a)	Correct repeat unit with continuation bonds (1)	1
127(a)	Correct repeat unit with continuation bonds (1)	1
128(a)	Correct repeat unit with continuation bonds (1)	1
129(a)	Correct repeat unit with continuation bonds (1)	1
130(a)	Correct repeat unit with continuation bonds (1)	1
131(a)	Correct repeat unit with continuation bonds (1)	1
132(a)	Correct repeat unit with continuation bonds (1)	1
133(a)	Correct repeat unit with continuation bonds (1)	1
134(a)	Correct repeat unit with continuation bonds (1)	1
135(a)	Correct repeat unit with continuation bonds (1)	1
136(a)	Correct repeat unit with continuation bonds (1)	1
137(a)	Correct repeat unit with continuation bonds (1)	1

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
138(a)	Correct repeat unit with continuation bonds (1)	1
139(a)	Correct repeat unit with continuation bonds (1)	1
140(a)	Correct repeat unit with continuation bonds (1)	1
141(a)	Correct repeat unit with continuation bonds (1)	1
142(a)	Correct repeat unit with continuation bonds (1)	1
143(a)	Correct repeat unit with continuation bonds (1)	1
144(a)	Correct repeat unit with continuation bonds (1)	1
145(a)	Correct repeat unit with continuation bonds (1)	1
146(a)	Correct structure (1); correct reagents (1)	2
147(a)	Correct structure (1); correct reagents (1)	2
148(a)	Correct structure (1); correct reagents (1)	2
149(a)	Correct structure (1); correct reagents (1)	2
150(a)	Correct structure (1); correct reagents (1)	2
151(a)	Correct structure (1); correct reagents (1)	2
152(a)	Correct structure (1); correct reagents (1)	2
153(a)	Correct structure (1); correct reagents (1)	2
154(a)	Correct structure (1); correct reagents (1)	2
155(a)	Correct structure (1); correct reagents (1)	2
156(a)	Correct structure (1); correct reagents (1)	2
157(a)	Correct structure (1); correct reagents (1)	2
158(a)	Correct structure (1); correct reagents (1)	2
159(a)	Correct structure (1); correct reagents (1)	2
160(a)	Correct structure (1); correct reagents (1)	2

Mark Scheme

MARK SCHEME — WEEK 3 (CONTINUED)

Q	MARKING POINTS	MARKS
161(a)	Correct structure (1); correct reagents (1)	2
162(a)	Correct structure (1); correct reagents (1)	2
163(a)	Correct structure (1); correct reagents (1)	2
164(a)	Correct structure (1); correct reagents (1)	2
165(a)	Correct structure (1); correct reagents (1)	2